Gateway Data Protocol of HTTP Communication

Document Revision History

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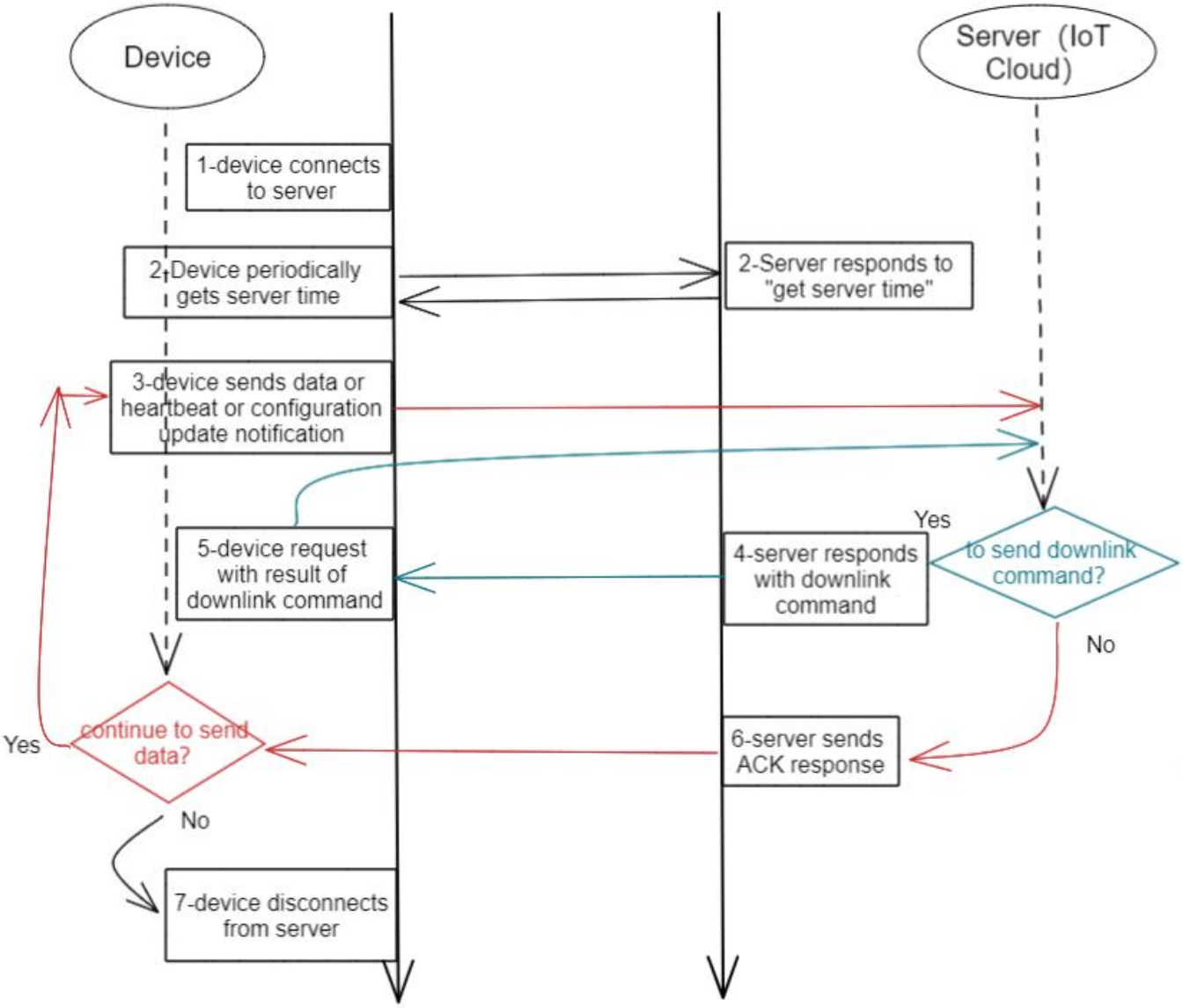
1. The Interaction Process and Related Requests and Responses

1.1 The Interaction Process Between the Server and the Device



1.2 Detailed Description of the Interaction Process

During http communication,

1. "1-Device connects to server": When the time interval is reached or triggered by other events, the gateway device connects to server.
2. "2-Device periodically gets server time": Device gets server time with a period like 6 hours which is different from the data time interval, so this request is not done every time after connecting to the server. After receiving the datetime in the response “2-Server Responds to get server time”, the device synchronizes its RTC time.

3. "3-Device sends data or heartbeat or configuration update notification": Gateway doesn't support heartbeat, so only data is sent.
4. The server judges the condition of "whether to send downlink command". After receiving sent data, the server checks the downlink commands queue to determine "whether to send downlink command". If yes, goes to "4-Server responds with downlink command "; if no, goes to "6-Server sends ACK response".
5. "4-Server responds with downlink command": The server takes the command at the top of the downlink commands queue and sends it to the device. These responses include the following 2 types:
   - Server Response with Downlink Command.
   - Server Response with 900 Downlink Command. (Only Supported by LoRa Gateway)
6. "5-device request with result of downlink command": After receiving and executing the command, the device responds to the server. These requests include the following 3 types:
   - Device's Request as Response to Downlink Command.
   - Device's Request as Response to 801 Downlink Command.
   - Device's Request as Response to 900 Downlink Command. (Only Supported by LoRa Gateway)
7. The server continues to judge the condition of "whether to send downlink command". If the judgment is yes, it cyclically executes steps 4-6 above. Exit when the judgment is no.
8. "6-Server sends ACK response": The server indicates that there is no commands to issue by sending ACK response, and hands over control to the device.
9. The device judges the condition of "continue to send data". The device checks the data-to-send queue to determine whether "continue to send data".
   - If yes, repeat "3-Device sends data or heartbeat or configuration update notification". Then it may enter the sub-loop of steps 4-6 above. After exiting the sub-loop, then to judge whether "continue to send data".
   - If no, execute "7-Device disconnects from server".
10. "7-Device disconnects from server". The device actively disconnects from the server, ending the communication between two sides. The server needs to wait for the next connection from the device to communicate. Therefore, when the server needs to send a downlink command to the device, it cannot issue it immediately and can only store it in the downlink command queue first.

1.3 General Description of Device Request and Server Response

1.3.1 Device Request

Only one http API with different payload is used for different requests.

http API: http://g-cloud.tzonedigital.com:18811/Receive (default). The URL can be changed by configuration software.

http method: post

request body: json data.

General example of request body:

```
{
   "msgtype":1,
   "hw":"0406",
   "fw":"02.02.00.00",
   "imei":"861234567891234",
   "rtc": “201012101500”,
   "sn":1,
   "data":{}
}
```

Payload description

Name	Value	Description
msgtype	Number	Message type, type of requests. value can be 1 to 6.
hw	Character string	Hardware model. 0406 for RD06/LoRa gateway.
fw	Character string	Firmware version like: 02.02.00.00
imei	Character string	Unique identifier. For gateway, it is the value of IMEI. For TT series, it is the value of device ID.
rtc	Character string, 12 characters	Format like “YYMMDDHHMMSS”, YY=Year of 20YY, MM=month, DD=day, HH=hour, MM=minute, SS=second Note: Not exist in every request.
sn	Number	Message ID (message serial number), from 1-9999 Note: Not exist in every request.
data	Json object	Data area with different sub-fields for different requests. Note: Not exist in every request. See payload description in the detailed description of requests

1.3.2 Server Response

Response body: json data.

General Example of response body:

```
{
    "sta": 0,
    "data": {},
    "error": "",
```

"errorcode":""

}

Payload description

Name	Value	Description
sta	Number	Status, 0 represents normal; 1 represents error
data	Json object	Data area with different sub-fields for different responses. Note: Not exist in every response. See "Special Symbol in Response Body" for details
error	Character string	The error message. See below.
errorcode	Character string	Error code. See below.

Error code and error message can be sent by server

Error Code	Error Message	Actions Taken by Device Accordingly
000	Unknown error	Disconnect from the server. Pause for 5 minutes and reconnect to the server
001	The server's receiving service is unavailable now.	Disconnect from the server. Pause for 5 minutes and reconnect to the server
002	The current number of connections exceeds the maximum number that the server can hold	Disconnect from the server. Pause for 5 minutes and reconnect to the server
003	The device sends data too fast	Disconnect from the server. Pause for 5 minutes and reconnect to the server. Note: people can change the data sending interval of the device if needed.
004	Data parsing error	discard the just-sent data and disconnect from the server. Pause for 5 minutes. Reconnect to the server and send the next data.
005	The RTC time sent by the device differs too much from the server time.	Discard all not-sent historical data and disconnect from the server. Pause for 5 minutes and reconnect to the server and sends get server time request.
006	Unknown device.	Disconnect from the server. Note: the error code means the server can't support the device hardware type in "hw" filed. People can shut down the device and contact the device supplier.
007	The device id is on the blacklist on the server-side.	Disconnect from the server. Pause for 5 minutes and reconnect to the server Note: This error occurs only when the server-side support blacklist feature. If it occurs, people can shut down the

		device or remove its device id from the blacklist on the server-side.

1.4 List of Device Requests

NO.	Device Request	Description	msgtype
1	Device Get Server Time request	Get Server Time to synchronize its RTC time.	1
2	Device Heartbeat request	Send heartbeat to server. Not supported yet.	2
3	Device Data Sending request	Device data, tag data records, etc.	3
4	Device's request as response to downlink command	General reply to downlink commands which are mainly related to the gateway device itself (except for instruction 900). For reading/query command, reply with query result; For setting command, reply with setting results	4
5	Device's request as response to 801 downlink command	Special reply to 801 downlink command which is related to the GSM version.	5
6	Device's request as response to 900 downlink command (Only Supported by LoRa Gateway)	Special reply to 900 downlink command which is related to the tags.	6

Note:

Some requests behave as response to server's downlink command.

1.5 A List of Server Responses

NO.	Server Response	Description	Special Symbol in Response Body
1	Server response to get server time	Return server time to device	"data" with "servertime" sub filed
2	Server response with downlink command	for setting or reading gateway device	"data" with "downcmd" sub filed
3	Server response with 900 downlink command (Only Supported by LoRa Gateway)	for setting or reading tags through gateway device	"data" with "downcmd" sub filed
4	Server ACK response	ACK response to show no downlink command to send, so device can decide what to do next.	No "data" field in the response body.

Note:

There is no “msgtype” field in the response body to distinguish different types of responses. Instead, special symbol is used in response body to show different types of responses

2. Detailed Description of All Requests or Responses

2.1 Device Get Server Time Request

Request:


```
{
   "msgtype": 1,
   "hw":"0406",
   "fw":"02.02.00.00",
   "imei":"861234567891234",
   "rtc": “201012101500”,
   "sn":1022,
}
```


Payload description

Name	Value	Description
msgtype	Number	Message type.1 for Device Get Server Time Request

2.2 Server response to “Get Server Time”

Response:


```
{
   "sta": 0,
   "data": {"servertime": “201012101516”
            },
   "error": ""
}
```


Payload description

Name	Value	Description
servertime	Character string, 12 characters	Format like “YYMMDDHHMMSS”, YY=Year of 20YY, MM=month, DD=day, HH=hour, MM=minute, SS=second

2.3 Device Heartbeat Request (Not supported yet)

Request:

{
"msgtype": 2,
"hw":"0406",
"fw":"02.02.00.00",
"imei":"861234567891234",
"rtc": “20201012101516”,
"sn":1022
}

Payload description

Name	Value	Description
msgtype	Number	Message type. 2 for Device Heartbeat Request

2.4 Device Data Sending Request

Request

{
"msgtype":3,"hw":"0406","fw":"02.02.00.00","imei":"865473033618318","rtc":"090101000737","sn":3,
"data":{
"mcc":"460","mnc":"01","lac":"2793","cell":"0FA1","alert":"AA","gsm":{"csq":26,"sta":"37"},"bat":4.09,"exvol":11.88,
"tag07b":[
{"id":"72180310","sta":"00","temp":-21.8,"humi":59,"rssi":-108,"bat":2.89,"rtc":"090101000637"},
{"id":"72170040","sta":"00","temp":22.1,"humi":67,"rssi":-111,"bat":3.67,"rtc":"090101000736"}
],
"tag09": [
{"id": “12345678”,"sta": “00”,"temp": -15,"rssi": -65,"bat": 3.6 ,"rtc":"181226201254"},
{"id": “81234567”,"sta": “00”,"temp": 25.2,"rssi": -61,"bat": 3.1,"rtc":"181226201254"}
],
"tag11": [
{"id": “12345678”,"sta": “00”,"voltage": 12.697, " current": 5.123," switch":0,"rssi": -65,"bat": 3.6 ,
"rtc":"181226201254"},
{"id": “81234567”,"sta": “00”,"voltage": 12.697, " current": 5.123," switch":0,"rssi": -61,"bat": 3.1,
"rtc":"181226201254"}
]
}
}

Note: The RD06 supports Tag06 series. The LoRa gateway supports Tag07/Tag08,/Tag09 and Tag11 series. The Tag06 series does not have an RTC time.

Payload description

Name	Value	Description
msgtype	Number	Message type. 3 for Device Data Sending Request
mcc	Character string	Mobile country code
mnc	Character string	Mobile network coding
lac	Character string	Area code
cell	Character string	Base station code
alert	Character string	Alert type, AA-normal data, 10-low voltage alarm, 60-external plug, 61-external disconnect
gsm	Json object	GSM profile
csq	Number	Signal strength（WIFI gateway is 00）
sta	Character string, 2 characters	Two hexadecimal digits for one byte to show some status. In the byte: （WIFI gateway is 00） bit5-Whether the http connected; 1-Http connected, 0-Http not connected; bit4-Whether the 4G network is ok; 1-4G connected, 0-4G network not connected; bit3-Whether to roam; 1-roaming, 0-not roaming; bit2-Whether the GSM network is ok; 1-GSM network connected, 0-GSM network not connected; bit1-Whether the SIM card is detected; 1-Detected SIM card, 0-No SIM card; bit0-Whether the GSM module is on; 1-GSM module is on, 0-GSM module is off;
bat	Number	battery voltage
exvol	Number	External voltage
tag06/tag07/tag08 /tag09/tag11	Json object	tag type name. It can be tag06/tag07/tag08/ tag09/tag11, etc. Different tag types can appear at the same time with data of their own. It is followed by array of tag data record. Note: The RD06 supports Tag06 series. The LoRa gateway supports Tag07, Tag08, Tag09, and Tag11 series
id	Character string	Tag device ID

sta	Character string, 2 characters	Two hexadecimal digits for one byte to show some status. In the byte: bit7: Battery voltage status, 1-low Voltage, 0- Voltage normal; bit6: Temperature alert status, 1-Temperature alert, 0- Temperature normal; bit5: Sensor button status, 1-Press sensor button 0-Don't press button button bit4: Sensor ACK switch 1-Sensor ACK enable; 0-Sensor ACK disable; bit3: RTC time; 1-Sensor RTC enable; 0-Sensor RTC disable; bit2-0:reserved;
temp	Number For other than tag11	Temperature.
rssi	Number	Signal strength.
bat	Number	Battery voltage.
rtc	Character string	Format like "YYMMDDHHMMSS", YY=Year of 20YY, MM=month, DD=day, HH=hour, MM=minute, SS=second
humi	Number	Humidity. for tag06B,tag07B,tag08B
temp2	Number	Temperature2. for tag09 with 2 sensors
voltage	Number	Voltage value, Unit: V. for tag11 only
current	Number	Current value, Unit: mA. for tag11 only
switch	Number	Switch value,0-connection,1-disconnect. for tag11 only

2.5 Server Response with Downlink Command

Response


```
{
    "sta": 0,
    "data": {
            "downcmd": "CMD-format"
            },
    "error":""
```

}

Note for CMD-format：

- Examples for CMD-format
  - Example 1 (Set data transmission interval of gateway device): *000000,018,10#

    So related response is: {"sta": 0, "data": {"downcmd": "*000000,018,10#"},"error":""}
  - Example 2 (Get/Read data transmission interval of gateway device): *000000,800,018#

    So related response is: {"sta": 0, "data": {"downcmd": "*000000,800,018#"},"error":""}
  - Example 3 (Get version of GSM module of gateway device): *000000,801#

    So related response is: {"sta": 0, "data": {"downcmd": "*000000,801#"},"error":""}
- For the CMD-format of specific instructions, please refer to the document "RD06/LoRa Gateway_4G/WIFI User Manual (command part)", and note the "Format" column, which is equivalent to CMD-format.
- CMD-format varies according to different instructions, in the following format:

  CMD-format=*{password},{ins-code}[,{ins-paras}]#
  - {} indicates variables, [] indicates optional.
  - password: password for accessing the gateway device, default is 000000.
  - ins-code: instruction code like 018, 800, 801, etc.
  - ins-paras: parameters of the instruction. For some instructions, no parameter is needed.
- Note instruction code 800 in the example 2. 800 is used to read value of any other instruction when instruction code is used as parameter, format like: *000000,800,{ins-code}#

2.6 Device's Request as Response to Downlink Command

Request


```
{
   "msgtype": 4,
   "hw":"0406",
   "fw": "02.02.00.00",
   "imei":"861234567891234",
   "data": {
         "resdowncmd": {
                        "cmd":"",
                        "cmdtype":"",
                        "sta":"",
                        "par":""
      }
       },
}
```


Payload description

Name	Value	Description

msgtype	Number	Message type. 4 for Device's Request as Response to Downlink Command
cmd	Character string	Received commands, such as*000000,008,0000001#
cmdtype	Character string	Command type; value can be “set” or “read”: set is to set command; read is to read command
sta	Character string	Set the command execution result. Value can be: OK or Err
par	Character string	Read command result

Examples:

- Example 1

  Server response with downlink command:

  {"sta":0,"data":{"downcmd":"*000000,008,0000001#"},"error":""}

  Device's request as response to downlink command:

  {"msgtype":4,"hw":"0406","fw":"02.02.00.00","imei":"866050033901164","data":{"resdowncmd":{"cmd":"*000000,008,0000001#","cmdtype":"set","sta":"OK"}}}

- Example 2

  Server response with downlink command:

  {"sta":0,"data":{"downcmd":"*000000,008,00000011#"},"error":""}

  Device's request as response to downlink command:

  {"msgtype":4,"hw":"0406","fw":"02.02.00.00","imei":"866050033901164","data":{"resdowncmd":{"cmd":"*000000,008,00000011#","cmdtype":"set","sta":"Err"}}}

- Example 3

  Server response with downlink command:

  {"sta": 0, "data": {"downcmd": "*000000,800,018#"},"error":""}

  Device's request as response to downlink command:

  {"msgtype":4,"hw":"0406","fw":"02.02.00.00","imei":"866050033901164","data":{"resdowncmd":{"cmd":"*000000,800,018#","cmdtype":"read","sta":"OK","par":"300"}}}

2.7 Device's Request as Response to 801 Downlink Command

Request

{
"msgtype": 5,
"hw":”0406”,

"fw": “02.02.00.00”,
"imei":"861234567891234",
"data": {
"gsmversion":"1418B05SIM800C24"
}
}

Payload description

Name	Value	Description
msgtype	Number	Message type. 5 for Device's Request as Response to 801 Downlink Command
gsmversion	Character string	version of GSM module of gateway device

Examples:

- Example 1

  Server response with downlink command:

  {"sta": 0, "data": {"downcmd": "*000000,801#"},"error":""}

  Device's request as response to downlink command:

  {"msgtype":5,"hw":"0406","fw":"02.02.00.00","imei":"865473033618318","data":{"gsmversion":"1418B05SIM800C24"}}

2.8 Server Response with 900 Downlink Command(Only Supported by LoRa Gateway)

Response

{
"sta": 0,
"data": {
"downcmd": "CMD-format"
},
"error":""
}

Note for CMD-format:

- Examples for CMD-format
  - Example 1 (Set TAG data transmission interval): *000000,900,12345678,002,5#

  So related response is: {"sta": 0, "data": {"downcmd": "*000000,900,12345678,002,5#"},"error":""}

  - Example 2 (Get/Read TAG data transmission interval): *000000,900,12345678,253,002#

  So related response is: {"sta": 0, "data": {"downcmd": "*000000,900,12345678,253,002#"},"error":""}

- CMD-format varies according to different instructions, in the following format:

CMD-format=*{password},900,{ID},{subCMD-format}#

- {} indicates variables, [] indicates optional.
- password: Password for accessing the gateway device, default is 000000.
- ID: device ID of tag, 8 characters.
- subCMD-format: Varies according to different sub-instructions. For details, please refer to the document "LoRa Gateway downward the command to TAG", and note the "instruction" and "Format" columns in the table. The former is sub-instruction, and the latter is equivalent to subCMD-format.
- Note sub-instruction code 253 in the example 2. 253 is used to read value of any sub-instruction for tag when other sub-instruction code is used as parameter, format like: *000000,900,12345678,253,{sub-ins-code}#

2.9 Device's Request as Response to 900 Downlink Command(Only Supported by LoRa Gateway)

Request


```
{
    "msgtype": 6,
    "hw":"0406",
    "fw": "02.02.00.00",
    "imei":"861234567891234",
    "data": {
        "tagcmd":{
                "id":"",
                "cmd":"",
                "cmdtype":"",
                "sta":"",
                "par":""
        }
}
```


Payload description

Name	Value	Description
msgtype	Number	Message type. 6 for Device's Request as Response to 900 Downlink Command
id	Character string	Device ID of tag
cmd	Character string	Instruction code for tag which is also called sub-instruction code.
cmdtype	Character string	Command type; value can be "set" or "read": set is to set command; read is to read command
sta	Character string	the command execution result. Value can be: OK or Err
par	Character string	Read command result

Examples:

- Example 1

  Server response with downlink command:

  {"sta":0,"data":{"downcmd":"*000000,900, 66660011,02,5#"},"error":""}

  Device's request as response to downlink command:

  {"msgtype":6,"hw":"0406","fw":"02.02.00.00","imei":"865473033618318","data":{"tagcmd":{"id":"66660011",
  "cmd":"2","cmdtype":"set","sta":"OK"}}}

- Example 2

  Server response with downlink command:

  {"sta":0,"data":{"downcmd":"*000000,900, 66660011,253,02#"},"error":""}

  Device's request as response to downlink command:

  {"msgtype":6,"hw":"0406","fw":"02.02.00.00","imei":"865473033618318","data":{"tagcmd":{"id":"6660011,
  ","cmd":"2","cmdtype":"read","par":"5"}}}

2.10 Server ACK Response

Response:

{
"sta": 0,
"error": ""
}

Usage:

- As response to device data sending request. If the server has no downlink command to send after receiving the request, it can send this response to the device
- As response to the 3 device requests responding to downlink command. If the server has no more downlink command to send after receiving the requests, it can send this response to the client.